What is a psychological assessment?

Psychological assessment is a series of tests conducted by a psychologist, to gather information about how people think, feel, behave and react. The findings are used to develop a report of the person's abilities and behavior—known as a psychological report—which is then used as a basis to make recommendations for the individual's treatment.
Psychological assessments and reports are used in other fields as well—like in the case of career planning for young adults or in the job application process to determine how well an applicant will fit into the position.

The procedures used to create an assessment are:

- Interviews
- Observation
- Written assessment
- Consultation with other mental health professionals
- Formal psychological tests

What is a psychological test?

A psychological test is used to measure an individual's different abilities, such as their aptitude in a particular field, cognitive functions like memory and intelligence or even traits e.g. introvert and extrovert etc. These tests are based on scientifically tested psychological theories.
The format of a test can vary from pencil and paper tasks to computer-based ones. They include activities such as puzzle-solving, drawing, logic problem solving, and memory games.

Some tests also use techniques—known as projective techniques—which aim to access the unconscious. In these instances, the subject's responses are analyzed through psychological interpretation and more complex procedures than the non-projective techniques mentioned above. For example, the Rorschach test, popularly known as the ink-blot test can provide insight into the person's personality and emotional functioning.

Psychological tests may also involve observing someone's interactions and behavior. Based on the result of the test, conclusion will be drawn about the individual's abilities and potential.

Difference between test and assessment

- A test is a measurement device or technique used to quantify behavior or aid in the understanding and prediction of behavior"
- Assessment includes more than just tests. A typical assessment plan may include a test or a battery of tests, interview, behavioral observation, and case history data

Uses of Psychological Test

1. **Understanding People**: They help understand how people think, feel, or behave.

2. **Comparing Differences:** These tests compare how individuals are different from each other.

3. **Measuring Changes:** They measure how someone might change over time in their thoughts or feelings.

4. **Assessing Abilities:** They check what someone is good at or what they might struggle with, like in school or work.

5. **Diagnosing Problems:** Psychological tests help identify issues like learning disabilities, emotional struggles, or mental health conditions.

6. **Guiding Decisions:** They assist professionals, like psychologists or counselors, in making decisions about treatment or support for someone.

7. **Assessment of Skills and Abilities:** Evaluating strengths and weaknesses in various areas, such as learning, memory, or problem-solving.

8. **Identifying Emotional Health:** Assessing emotions and mental health conditions like anxiety, depression, or stress.

9. **Measuring Personality Traits:** Analyzing different personality aspects such as introversion/extroversion, conscientiousness, or openness to experience.

10. **Diagnosing Disorders and Conditions:** Detecting conditions like ADHD, autism, or specific learning disabilities.

11. **Research and Development:** Contributing to research by providing data and insights into human behavior, cognition, and emotions for developing new theories or treatments.

Types of Psychological tests

1. Individual versus group tests
2. Ability, achievement, or aptitude tests
3. Intelligence versus personality tests
4. Speed test versus ability tests
5. Structured personality tests versus projective tests
6. Verbal versus non-verbal/performance tests.
7. Commercial copyrighted tests versus available to all test

Individual versus Group Tests:

Some tests are for one person at a time (like WAIS, WISC), while others can be done with many people together (like Raven's Progressive Matrices). Individual tests need one examiner for one person, while group tests are quicker, with multiple people and often use multiple-choice questions. Group tests are easier to score and can be administered through recorded instructions or computers. However, they might not be ideal when a good connection between the person being tested and the examiner is necessary.

Ability, achievement, or aptitude tests

Ability tests, like intelligence tests, measure your thinking and learning skills, providing an IQ score. They show what you know at a certain time and your level of skills in different areas.

Achievement tests check what you've learned from specific training or programs, like SATs at the end of courses, to see if you met the goals.

Aptitude tests measure the overall effect of your life experiences on your learning abilities. They predict how well you might do in the future, unrelated to specific training. Differential Aptitude Tests (DAT) are common examples.

Intelligence Verses Personality Tests:

Even though the test names suggest what they measure, it's important to know that personality tests don't tell you about IQ, and IQ tests can't tell you about personality traits. Many confuse these, but they're different and come in many varieties, both being widely available**.**

Speed Tests versus Ability Tests:

In ability tests, like IQ tests, the focus is on measuring the level of ability, not how fast you finish. Some tests, such as Raven’s Progressive Matrices, don’t have a time limit. However, in other tests like Clerical Speed and Accuracy Test, how quickly you perform is important. Power tests have a time limit, but it's long enough for everyone to finish.

Structured Personality Tests versus Projective Tests:

Personality tests come in different types and measure usual behavior like traits. There are two main categories: structured tests and projective tests. Structured tests have fixed answer options, like true/false or multiple-choice questions, where you pick the one that best describes you. These tests are often self-report type.

Projective tests, unlike structured ones, are harder to score and interpret. They involve showing vague images or words to people who then describe what they see or complete the task. For example, Rorschach’s Inkblots or TAT, where people tell stories about pictures, are used to understand personality. Other tests, like HTP or RISB, involve drawing or completing sentences to reveal traits or tendencies.

Verbal versus Nonverbal/ Performance Tests:

Many tests rely on language skills, especially IQ tests and structured personality tests. However, tests like Raven’s Progressive Matrices don’t measure verbal abilities or use language.

Commercial-Copyrighted Tests Versus “Available To All”/Online Tests:

Most copyrighted IQ tests, like WAIS or WISC, can't be copied or used without permission as it's unethical and diminishes their value if people already know the content. However, some tests available online or in books are mainly for research, not for diagnosing or screening. Personality tests are more commonly found online, like the Multidimensional Health Locus of Control Scale or Self Efficacy Scale, which authors have made available for use, downloading, and printing, offering valuable research evidence**.**

Ethical, Legal and Professional issues in assessment

1. **Confidentiality and Privacy:** Safeguarding sensitive information collected during assessments to ensure it remains private and is shared only with authorized individuals or for authorized purposes.

2. **Informed Consent:** Obtaining explicit consent from participants or individuals being assessed, providing comprehensive information about the assessment process, its objectives, and how the results will be used.

3. **Fairness and Equity:** Ensuring assessments are conducted impartially, free from bias or discrimination based on factors like race, gender, ethnicity, or socioeconomic status.

4. **Validity and Reliability:** Ensuring assessment tools accurately measure what they are intended to measure and produce consistent, dependable results.

5. **Professional Competence:** Conducting assessments within one's area of expertise, adhering to professional standards, and maintaining ongoing professional development.

6. **Legal Compliance:** Abiding by laws and regulations governing assessments, such as those related to data protection, disability accommodations, and informed consent.

7. **Avoiding Conflicts of Interest:** Preventing situations where personal interests or biases could influence assessment outcomes, ensuring objectivity and fairness.

8. **Appropriate Use of Results:** Using assessment outcomes responsibly, ensuring accurate interpretation, and avoiding misuse that could have adverse consequences for individuals.

9. **Cultural Sensitivity:** Recognizing and addressing cultural influences that might affect the assessment process or interpretation of results, ensuring assessments are culturally relevant and unbiased.

10. **Clear Communication and Feedback:** Providing understandable and meaningful feedback to individuals being assessed, ensuring they comprehend the results and their implications for further action or support.

Adherence to these ethical, legal, and professional principles is vital in maintaining integrity, fairness, and respect for individuals throughout the assessment process.

Characteristics of a Test

- **Reliability**

What Is Reliability in Psychology?

According to Kaplan & Saccuzzo (2001) reliability is "the extent to which a score or measure is free from measurement error. Theoretically, reliability is the ratio of true score variance to observed score variance."

Measurement Error

The classical test score theory implies that everyone can obtain a true score on any test or measure if the measure is free of error. But we know that there is perhaps no measure that can be considered as totally error free. There is always some chance of error. This further implies that the scores that we obtain on different measures are not the true scores of the test taker. These are the observed scores plus error. In other words the observed score is not the true score of a test taker, implying that the observed score may not be a 100% true representative of a person's traits or abilities. Here we need to realize that the term 'error' is not used to indicate that something has gone wrong or we have made some 'mistake'. Error, in this context, refers to the amount and extent of variance that may be expected in results. The observed score is therefore the sum of true score and error i.e.

$X= T+E$

Here X refers to the observed score, T is the true score, and E is the error that can be expected in the measurement

Sources of Error in Test Scores: Psychologists and educationists try to estimate the amount and degree of error that may be expected in their measurement. That is the main reason why test developers and test administrators emphasize on uniform testing procedures besides controlling other possible sources of error.

The following are a few of the variables that may be possibly causing error in measurement:

A. Test Related Factors:
    - Difficulty level (too difficult or too easy)
    - Length of the test (too long causing fatigue or boredom)
    - Domain or content (not suitable for all test takers…..suitable for some but not all)
    - Items may not be representing the domain or content
    - Time limit (may cause stress or a handicap)

B. Test Administration Process:
    - Poor and not uniform testing conditions (physical setting and environment)
    - Poor, faulty, improperly worded, improperly delivered, not uniform instructions

- Test administrator's personality (different administrators in different situations having different personality styles)
- Rapport (poor, too much, or too little)

C. Examinee Related Variables:

- Prior learning and experience
- Individual differences (within group differences; Personality styles, stress tolerance level, IQ level, emotionality, motivation, knowledge)
- Difference from the normative sample (no group is identical to the normative sample; every group is different from every other group)
- Within- person differences (the same persons may change over time (life, academic, and professional experiences; physical conditions, health, motivational level, emotional state etc.)

Correlational and reliability

Correlation shows the relation between 2 variables, and its measure by a statistics called the correlation of coefficient.

- Coefficient of correlation tells us two things about a relationship; the magnitude of relationship and the direction of relationship. Magnitude means the size of correlation whereas direction indicates whether the correlation is positive or negative.
- The size of correlation range between -1 (perfect negative) and +1(perfect positive) with a zero (no correlation at all) value in between
- The value of coefficient of a correlation is always 1 or less than 1
- The closer is a value of correlation to one, the stronger is the correlation e.g. r= 0.9.
- The closer a coefficient of correlation is to zero, the weaker will be the relationship.

Types of Reliability

According to the definition given by Anastasi & Urbina (2007) reliability refers to "the consistency of the scores obtained by the same persons when they are reexamined with the same test on different occasions or with different sets of equivalent items, or under other variable examining conditions". There are two main types of reliability: internal reliability and external reliability.

1. **N Internal reliability** means that the measure has consistency within itself. In other words, the same question posed differently would produce the same results. It is often measured using the split-half method.

2. **External reliability** refers to how results compare to results between individuals and across time. It is often measured using test-retest, inter-rater, and parallel forms methods.

Measurement of Reliability

Test-Retest Reliability

Test-retest reliability is a measure of the consistency of a psychological test or assessment. This kind of reliability is used to determine the consistency of a test across time. Test-retest reliability is best used for things that are stable over time, such as intelligence.

Test-retest reliability is measured by administering a test twice at two different points in time. This type of reliability assumes that there will be no change in the quality or construct being measured. In most cases, reliability will be higher when little time has passed between tests.

The test-retest method is just one of the ways that can be used to determine the reliability of a measurement. Other techniques that can be used include inter-rater reliability, internal consistency, and parallel-forms reliability.

A major **advantage** of this method is that there is maximum control over the test taker and test item variables. The subjects are the same and the test items are also the same.

Drawbacks

1. **Changing Conditions**: Sometimes, the environment or circumstances might change between two tests. Things like noise or temperature can affect how well someone performs.

2. **Changes in People:** Test takers might change too. Their feelings, motivation, social situations, or even their health might be different the second time they take the test.

3. **Practice Effect:** Doing the same test twice might make it less challenging or more familiar for the person. Or it might become boring. This can affect their performance.

4. **Learning from the Test:** Some tests are such that people might remember the questions or figure out how to answer them better the second time. This can especially affect tests that measure thinking skills like problem-solving, reasoning, or math.

5. **Time Gap Matters**: The time between the two tests is important. If the gap is short, the results might be more similar. But if it's longer, the results might differ more. Test makers should mention the time gap when they talk about how reliable the test

Test- retest reliability is denoted as **rtt.**

Inter-Rater Reliability

This type of reliability is assessed by having two or more independent judges score the test. The scores are then compared to determine the consistency of the examiners estimates.

One way to test inter-rater reliability is to have each examiner assign each test item a score. For example, each examiner might score items on a scale from 1 to 10. Next, you would calculate the correlation between the two ratings to determine the level of inter-rater reliability.

Another means of testing inter-rater reliability is to have examiners determine which category each observation falls into and then calculate the percentage of agreement between the examiners. So, if the examiners agree 8 out of 10 times, the test has an 80% inter-rater reliability rate.

Parallel-Forms Reliability / Alternate-Form

Parallel-forms reliability is assessed by comparing two different tests that were created using the same content. This is accomplished by creating a large pool of test items that measure the same quality and then randomly dividing the items into two separate tests. The two tests should then be administered to the same subjects at the same time.

Independently developed and completely parallel or equivalent forms of the same measure. They should match each other in all respects including:

- Same specifications
- Same instructions
- Same time limit
- Same content
- Same number of items
- Same item format
- Difficulty level

Consistently and stability across the parallel of forms

Consistency over time

	Form A	Form B
Occasion 1	Group I (50% of total)	Group 2 (50% of total)
	Interval	
Occasion 2	Group 2 (50% of total)	Group I (50% of total)

Advantages

1. Increases Confidence: Using two different tests boosts confidence in the accuracy of results.
2. Minimizes Memory Effects: Prevents people from recalling answers between tests, ensuring more reliable results.

3. Provides Diverse Insights: Different tests offer a broader understanding of someone's abilities.
4. Identifies Test Issues: Consistent results indicate a reliable test, while differences might signal problems.
5. Promotes Fairness: Using different forms ensures fairness without favoritism for specific questions or formats.
6. Enhances Flexibility: Having various test versions can be advantageous in different situations.

Drawbacks

- Hard to Create: Making two tests that are equally good at measuring the same thing is tough. They need to be equally challenging and cover the same topics.
- Difficult to Manage: Giving two different tests to the same group can be tough to organize, especially in big groups or when testing conditions need to be the same.
- Small Differences: Even if we try hard, there might be tiny differences between the two tests that affect how well they measure the same thing.
- Not Always Possible: Some things are hard to test with two different versions. For example, certain skills or abstract ideas might not easily work with different test forms.
- Practice Effect: If people get used to the type of questions on the first test, they might do better on the second test just because they're more familiar with it. This can mess up the reliability of the test.
- Takes Time and Money: Creating and checking two versions of a test needs a lot of time, effort, and money. This might not be possible or practical in some situations.

Split half Method Reliability

This form of reliability is used to judge the consistency of results across items on the same test.

Essentially, you are comparing test items that measure the same construct to determine the tests internal consistency. It is often referred to as the split-half method of measuring reliability.

When you see a question that seems very similar to another test question, it may indicate that the two questions are being used to assess reliability.

Because the two questions are similar and designed to measure the same thing, the test taker should answer both questions the same, which would indicate that the test has internal consistency.

The test can be divided in many ways:

- Divide test into equal number of items.

The first 50% items are considered as one half and the remaining 50% as the other half. Scores are calculated separately for the two halves. There is one limitation of dividing the test this way. There is likelihood that the two halves may be having different difficulty levels. The initial items may be easier than the items in the second half. Also, items in the two halves formed in this way may pertain to different content areas

- A better option can be to consider odd numbered items in one half, and the even numbered items in the other half.

This formula makes it possible to estimate the internal consistency reliability of the test from the reliability of the two halves of the test. The formula is used for the computation of reliability while taking care of the fact that a test has been shortened or lengthened. The formula also considers the number of times the test has been shortened or lengthened.

$$r_{nn}=\frac{nr_{tt}}{1+(n-1)r_{tt}}$$

Spearman- Brown formula is given below:

rnn= estimated coefficient

rtt= obtained coefficient

n= number of times the test has been lengthened or shortened e.g. two times, three times etc

In split- half reliability the test is reduced to half, therefore the reliability of the full test is calculated using the Spearman- Brown formula that involves doubling the length of the test, as follows:

$$r_{tt}=\frac{2r_{hh}}{1+2r_{hh}}$$

Draw backs

1. **Splitting Impact: Reliability** varies based on how the test is split**.**
2. **No Item-Level Assessment:** Doesn't gauge consistency within individual questions.
3. **Small Group Challenges**: Less accurate with fewer test-takers.
4. **Accuracy Risks:** Short tests may give unreliable reliability estimates**.**
5. **Assumption of Equal Difficulty:** Assumes both halves are equally challenging.
6. **Chance Influence:** Splits may sometimes result in random outcomes.
7. **Limited for Diverse Tests:** Not ideal for tests with varied question types**.**
8. **Length Influence:** Longer tests tend to offer more reliable estimates.
9. **Mixed Subject Disadvantage:** Might not accurately reflect overall consistency in tests covering different subjects.

Kuder- Richardson Reliability:

The Kuder-Richardson method is somewhat like the split-half method because both check how reliable a test is using just one administration. But, they're different in how they look at consistency. Instead of splitting the test into halves, the Kuder-Richardson method looks at how each question performs. Basically, it's the average of all the split-half reliability scores you can get from a test.

$$r_{tt=}\left(\frac{n}{n-1}\right)\frac{SD_t^2-\sum pq}{SD_t^2}$$

r_{tt}= Coefficient of reliability of whole test
n= the number of items in the test
SD_t^2= Standard deviation of total scores on the test is SD_t, squared (=variance)
p= proportion of persons who pass each item
q= proportion of persons who do not pass each item
pq= product of (p) and (q) for each item
$\sum pq$= The sum of products of (p) and (q) for all item

- Only used for that items that their answers are either right or wrong / zero or one
- Cannot use with Likert scale and having weightage

Coefficient Alpha:

Coefficient alpha is another method, much like Kuder-Richardson, used to check how consistent a test is. But while Kuder-Richardson works only for tests where answers are either right or wrong (like true/false), coefficient alpha works for tests where answers get different scores, like in personality tests where answers have different weights. This formula helps measure the consistency of these types of tests.

$$r_{tt=} \left(\frac{n}{n-1}\right)\frac{SD_t^2 - \sum(SD_i^2)}{SD_t^2}$$

r_{tt}= Coefficient of reliability of whole test

SD_t^2= Standard deviation of total scores on the test is SD_t, squared (=variance)

SD_i^2 = Variances for item scores

$\sum(SD_i^2)$= sum of the variances of item scores

Standard error of measurement

The standard error of measurement (SEm) estimates how repeated measures of a person on the same instrument tend to be distributed around his or her "true" score. The true score is always an unknown because no measure can be constructed that provides a perfect reflection of the true score.

The Standard Error of Measurement (SEM) is a measure of how much mistakes or variability exists in a test. It's like a measure of the "ups and downs" in test scores for the same person taking the test multiple times.

To figure out SEM, you use a formula. If you know the reliability of the test (how consistent it is), and the standard deviation of the test scores, you can calculate SEM using a simple formula. The formula uses the test's reliability and the variation in scores to estimate the amount of error or variability in the test results.

$$SEM = SD_t\sqrt{1 - r_{tt}}$$

SDt = Standard deviation of the test score

Rtt = reliability coefficient

For example, if a test had a standard deviation of 12 and the reliability coefficient is .90, then SEM will be= 12√ 1- .9= 3.79 .The SEM is the standard deviation of the hypothetical distribution. If XYZ attains an average IQ 0f 90 in the hypothetical situation it will be her true score that may range between 90- 3.79 and 90+ 3.79.we remember that in normal distribution about 68% people fall between -1 and +1 standard deviations from the men. Therefore applying the same concept we can say that 68% people may be expected to score within this range of scores.

Validity

Traditionally, the validity of a test has been defined as the extent to which a test measures what it was designed to measure.

What a test measures and how well it measures

The extent to which a test measures the quality it purports to measure. Types of validity evidence include content validity, criterion validity, and construct validity evidence

Psychological tests are developed primarily for two reasons:

The validity of a test should not be taken casually, and cannot be reported in general terms either. Validity is not, and should not be measured and reported as “highly” valid or “low” validity. It is reported in specific terms.

- Relationships between the test content (items) and the domain (area) that it represents
- Relationship between the test content and the objectives (purpose) that lead to test construction
- Relationship between performance (r/s) and performance on other independent measures that are known to measure the same phenomenon/trait/ability/domain etc

Psychological tests are developed primarily for two reasons:

- Achievement testing: In order to see if a specific content area has been learnt and/ or mastery is acquired, and
- Predictive function: predicting future performance on the basis of the present test’s performance .

Measurement of validity

There are four type of validity

1. Construct validity
2. Content validity
3. Face validity
4. Criterion validity

- **Content Validity Evidence:**

This process is about figuring out what a test really measures by doing different studies. The researcher defines what they want to measure and creates a test for it. By looking at how well the test matches up with other things, like other tests or measurements, they can understand what the test actually means.

- **Construct Validity**

A process used to establish the meaning of a test through a series of studies. To evaluate evidence for construct validity, a researcher simultaneously defines some construct and develops the instrumentation to measure it. In the studies, observed correlations between the test and other measures provide evidence for the meaning of the test

- **Criterion validity**

"The evidence that a test score corresponds to an accurate measure of interest. The measure of interest is called the criterion

Content Validity

- A systematic examination of the test content is made in order to determine whether it covers a representative sample of the behavior domain to be measured.
- Such validation procedures measure how well the individual has mastered a specific skill or course of study

Content validity is built into a test from the outset through the choice of appropriate items. For educational tests, the preparation of is preceded by a thorough and systematic examination of relevant course syllabi and textbooks, as well as consultation with subject matter experts. On the basis of the information thus gathered, test specifications are drawn up for the item writers

Certain considerations are needed for content validation measure:

- The test shod cover all major aspects and items in correct proportion (the overloading and underrepresentation of elements should be avoided).
- The domain under consideration must be fully described in advance rather than after test preparation.

- Content should be broadly covering the major objectives, application and interpretation and also factual knowledge

Specific Procedures:

The choice of appropriate items is important in content validity. A number of careful decisions are taken at the time of test development e.g

Appropriate selection, appropriate formation, appropriate wording, appropriate order of the items

For educational tests, items are prepared form relevant course syllabi and textbooks and in some cases consultation with subject experts is used

Test specifications are given to item writers

- It includes instructional objectives, relative importance of topics/processes.
- It should clearly indicate the number of items for each topic.
- It may also provide sample material.
- The process of content validation should include description of all procedures in the manual; that ensure that test is appropriate and representative.

For example in subject-specialists has participated, the names, qualification and number of individuals should be stated.

The test total score and achievement score of an individual can be checked for grade progress

Application

- Content validity is basic to the validity of educational and occupational achievement tests.
- This validity measure is applicable to occupational tests designed for employee selection and classification.
- For aptitude and personality test content validation in inappropriate.

Content Validity versus Face Validity

We should not confuse content validity with face validity. Many students of psychology take them to be one and the same, whereas actually they are different.

Face validity is about the test appearing to be valid. It is not validity in the technical sense; nor is it technically measured.

Face validity is defined as "The extent to which items on a test appear to be meaningful and relevant, actually not evidence for validity because face validity is not a basis for inference" (Aiken, 1994, p.636). Face validity may be used where prediction r inference are not involved e.g. questionnaire or scale for a survey**.**

Criterion Validity

Whenever we plan and design a test we have a certain standard in mind that we want to meet.

- To look for a correlation between the test and another related test.
- We compare the results of our test with the results / score of another test which we think is a standard that we want to achieve, which is a standard that we want to compare with

Criterion- related validity refers to a procedure where scores o a test being used are correlated with scores on a criterion

The evidence that a test score corresponds to an accurate measure of interest. The measure of interest is called the criterion" (Kaplan & Saccuzzo, 2001, p.635).

Why do we need a criterion to estimate the validity of our measure? To understand this concept let's look at two simple everyday example

In other words, for example, if a teacher develops a test of mathematical ability, how is she going to determine its validity? How will she judge that her test is measuring mathematical ability and nothing else? Most probably she will try to examine the relationship between scores on her test with another authentic test of the same ability, or may be a test of similar skills. By doing so, she will be assessing the criterion- related validity of her test.

Criterion Prediction Procedures:

- **Predictive validity evidence**

"The evidence that a test forecasts score on the criterion at some future time" (Kaplan & Saccuzzo, 2001, p.638). Predictive validity pertains (related to) to prediction over a time interval. Predictive validation yields information that is most relevant to tests that are used for predicting future performance of an individual.

Predict over a time interval (latter sense)

There are situations where tests are used to classify, select, and even reject individuals.

For example tests are used for selecting students for an academic program, for personnel selection and job placement, or for choosing staff members for a specific training or higher education.

Similarly there are times where rather than being used for selection; tests are used for screening out (filter) individuals. The tests are used to predict if the candidate is likely to develop an undesirable behavior, characteristic, mental or emotional problem etc. in future.

- **Concurrent validity evidence**

Concurrent validity is relevant to tests employed for diagnose existing status rather than prediction of the future

In this approach the test is administered to such a sample whose data on the criterion are already available

Concurrent valid evidence is defined as "evidence for criterion validity in which the test and the criterion are administered at the same point in time" (Kaplan & Saccuzzo, 2001, p.635).

When the validity of a test is measured with reference to a criterion, and both are administered at nearly the same time, then the resulting validity will be concurrent validity

The primary idea is whether this test can differentiate between people with specific characteristics. If the test can do so, then it can be used as an efficient tool for categorizing people according certain criteria or classifications. One of the best examples of such a tool can be diagnostic tests e.g. Minnesota Multiphasic Personality Inventory (MMPI).

The criterion data are always available at the time of testing.

"The logical distinction between predictive and concurrent validation is based not on time but on the objectives of testing. Concurrent validation is relevant to tests employed for diagnosis of existing status rather than prediction of future outcomes

Example

Does smith qualify as a satisfactory pilot? (Concurrent validity)

Does smith have the qualities to become a satisfactory pilot? (Predictive validity)

Criterion Measures

There is no limit to the type of criterion that may be used for the purpose of test validation. However some are used more frequently and popularly than others. Tests may involve a number of criteria. Some of the criteria are mentioned here**:**

a) Academic achievement; used for intellectual tests and similar tests
b) Matriculation marks at the time of admission to higher classes; used for selection and screening.
c) For those who have completed, or discontinued education, years of education can be good criteria.
d) Performance in specialized training; used for selection, admission, achievement.
e) Instructor's ratings
f) Grades in a semester

Characteristics of a Criterion:

Reliable; the scores on the criterion as well as those on the test should be reliable. The reliability coefficients of the two sets of scores affect the coefficient of validity. This can be understood from the following pattern of relationship

$$r_{xy} \leq \sqrt{(r_{xx})\left(r_{yy}\right)}$$

Rxy stands for the coefficient of validity

test reliability and criterion reliability are represented by rxx and r yy. It can be seen that the two coefficients of reliability contribute to the validity coefficient. Weak/little/or no reliability will automatically affect validity.

Valid; it should measure what it is supposed to measure.

Relevant; it should be relevant to the purpose for which it is being used.

Uncontaminated; (pollute) we should try to use uncontaminated criterion. There are situations when the criterion itself has been, partially or fully, based on predictor measures. This is called criterion contamination.

They should have similar things, similar trait, similar characteristics they may be based upon similar ideas but they should be independent of each other at the same time (same items)

Construct Validity

Construct validation has focused attention on the role of theory in test construction and on the need of formulation hypotheses that can be proved or disapproved in the validation process.

Some years back I developed a test for the assessment of science concept acquisition, called SCAT. The purpose of developing this test was to develop a tool that could measure the extent to which children in the final year of their school have learnt the application of science concepts that they were taught in the past 4-5 years.

The reliability and validity of the test were also measured. In order to see if the test was valid or not, three assumptions were formulated:

- The SCAT scores will have a high-positive correlation with scores on a test of similar nature and content. (similar nature)
- The SCAT scores will have a high-positive correlation with scores on a test of reasoning ability.(similar ability)
- The SCAT scores will have a moderate-positive correlation with marks on school science achievement test.
- The SCAT scores will have either no or a negative correlation with scores on a test of rote memorization of meaningless material like non-sense syllables

In order to test these four assumptions, four different tests were used for these assumptions respectively:

- Dallas Times Herald Test (scale test) (science test)
- Basic Reasoning skills Test
- General Science Test used in school's annual examination,
- list of non-sense syllable

Construct Validity Evidence:

"Construct validity is a judgment about the appropriateness of inferences drawn from test scores regarding individual standings on a variable called a 'construct'. A construct is an informed scientific idea developed or constructed to describe or explain behavior".

"Constructs are unobservable, presupposed (underlying) traits that a test developer may invoke to describe test behavior or criterion performance" (Cohen, 1999)

The significant elements of this description are as follows: (steps)

- A theoretical construct or trait.
- Each construct is developed to explain and organize observed response consistencies.
- It derives from established interrelationships among behavioral measures. (tool)

- Requires the gradual accumulation of information from a variety of sources

Assumptions Underlying Construct Validity:

- Test scores will be highly correlated with tests measuring same construct/similar construct. This is **convergent validity.**
- Test scores will have weak/low (or at times may be negative) correlation with tests meant to measure construct which are different from the one measured by the main test. This is **discriminant validity**. (covers a different construct)

Convergent Validity Evidence: This aspect of validity shows us the extent to which a test measures the same construct/trait/ attribute as do other measures that have been designed and developed to measure the same construct/trait/ attribute

Discriminant Validity Evidence: Discriminant validity provides us information that the test in question does not measure what other tests measure, or it measures something different from what other available tests measure

Can we measure both validity aside?

Because they support each other validity

Steps in Construct Validity Process:

- The specific test being referred to must be present.
- Clearly defining the construct to be examined
- Assumptions about the construct
- Assumptions about the similar constructs and the different constructs
- Identification of measures (tests) that will be used to test the constructs in question
- Test administration, may be in various steps (session , situation)
- Calculation of correlations between the test scores
- Analysis of correlations in order to reach final conclusions regarding construct validity